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PALAEOMAGNETIC RESULTS ON THE DECCAN TRAP LAVAS OF THE AURANGABAD REGION AND THEIR TECTONIC SIGNIFICANCE

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ABSTRACT

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Detailed palaeomagnetic measurements have been carried out on a 600 m sequence of Deccan trap lava flows sampled in a 300 km long, NW–SE traverse through the trap country between Aurangabad (19°51'N 75°16'E) and Diglur town (19°48'N 77°36'E) which is located on the eastern margin of the traps. From 28 basalt flows between altitudes of 344 m to 927 m were collected 175 oriented block samples. Stable remanent magnetic directions obtained after alternating field demagnetisation tests indicate that all the lava flows are reversely magnetised. The mean value for the remanent directions was found to be: azimuth 150° and inclination 48° downwards; the palaeomagnetic pole position corresponding to this direction is 33°N 73°W.

Earlier palaeomagnetic studies had revealed the existence of polarity transition from reverse to normal magnetisation at an approximate altitude of 600 m at several locations in the Deccan trap sequences on the western coast. The absence of this polarity transition record in the trap sequence of the Aurangabad region suggests that the two sectors might have had a different eruption or tectonic history. Available data on the Bouguer anomaly patterns over the Deccan trap, favour the interpretation suggesting that the Aurangabad region had been subsequently uplifted by at least 325 m.

INTRODUCTION

The Deccan trap represents the largest single unit of plateau basalts on earth. At present they are spread over an area of about 512,000 km^2 (Fig.1) and their original extent is believed to be around 1,280,000 km^2. They attain a maximum thickness of about 1800 m near the western coast. On the basis of stratigraphic evidence, the Deccan trap is considered to have started erupting sometime near the end of the Cretaceous and continued till Early Eocene (Wadia, 1961). Recent measurements of K–Ar whole rock ages of four basalt samples have led Wellman and McElhinny (1970) to suggest that the actual span of this activity is of the order of 5 m.y., the average age being around 60 m.y.

Palaeomagnetic studies have been carried out on many localities in the Sahyadri ridges running parallel to the western coast (Clegg et al., 1956; Deutsch et al., 1958, 1959; Sahasrabudhe, 1963; Wensink and Klootwijk, 1971). The palaeomagnetism of the Deccan

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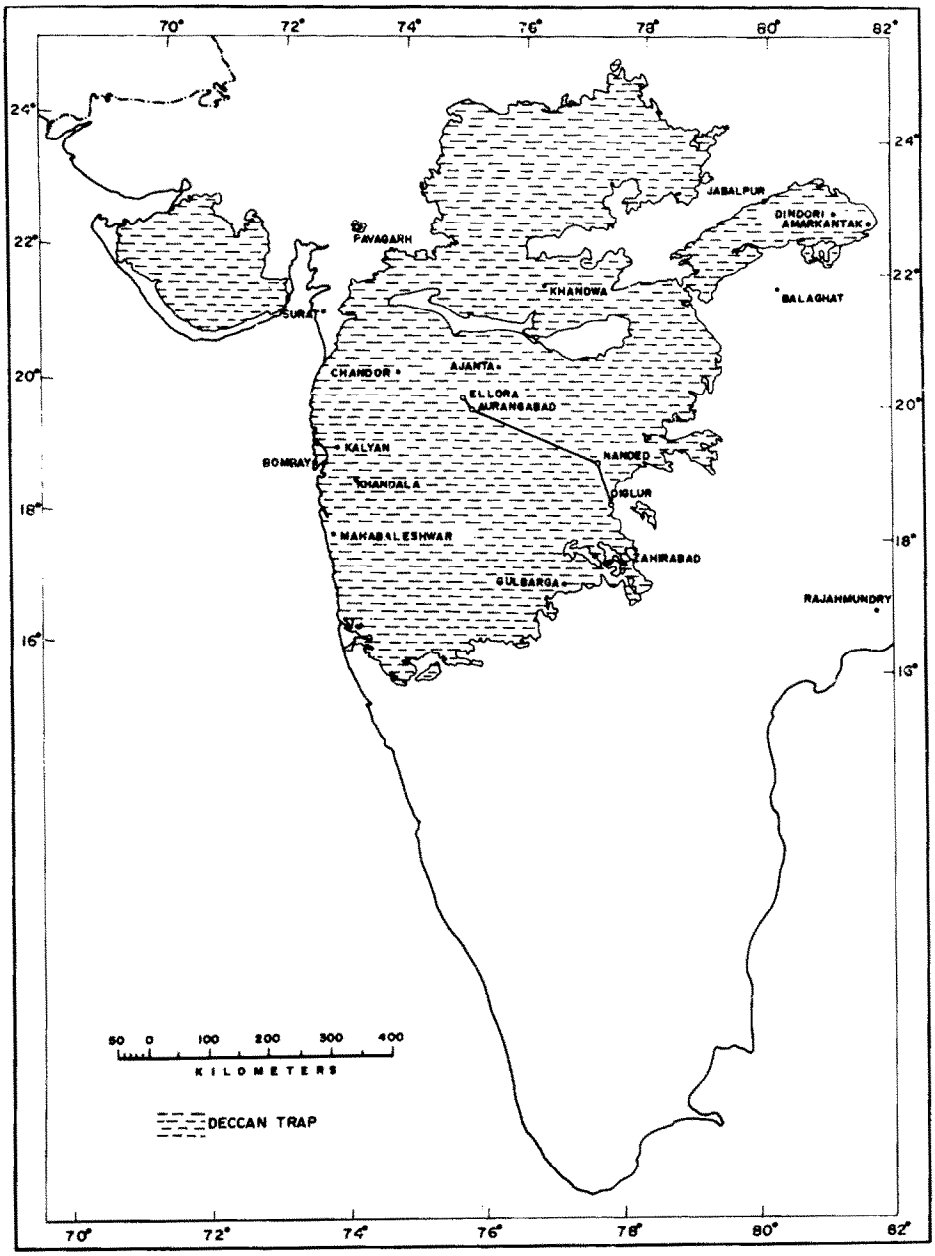


Fig.1. Map of Peninsular India showing Deccan trap outcrops.

trap near Jabalpur (Verma and Pullaiah, 1968) and Amarkantak (Athavale, 1970) on the northeast side, and of trap flows near Gulbarga (Verma and Apparao, 1963) has also been studied. The following broad conclusions have emerged as a consequence of these studies:

(1) During the eruption of the Deccan lava sequence, India was situated in the Southern Hemisphere, and was drifting northwards.

(2) During the eruption of the Deccan trap, the earth's magnetic field underwent a reversal. Thus we get two polarities in the trap flows. Flows below an approximate altitude of 600 ± 60 m are reversely magnetised, while those above this altitude are normally magnetised. This observation was made in the western ridges at several sites (Deutsch et al., 1959) and has been found to hold good (Verma and Pullaiah, 1968) in some far-off locations in the Deccan trap, such as the lava sequence near Jabalpur (see Fig. 1).

It is reasonable to assume that the general conclusions drawn by Sahasrabudhe (1963) that the Deccan trap could be classified into two groups: (1) Lower (reversed), and (2) Upper (normal) on the basis of polarity of magnetisation, is valid for most of the Deccan trap-covered area.

DETAILS OF SAMPLING

In this paper we report the results of palaeomagnetic studies carried out on a sequence of Deccan trap lava flows, exposed on the eastern side and ranging in altitude from 344 m to 927 m. In a 300 km long, SE–NW traverse (see Fig.1), 28 lava flows were sampled between Diglur (19°48'N 77°36'E) and Ellora (20°01'N 75°11'E) near Aurangabad. Diglur is located on the eastern margin of the Deccan trap. The unconformable junction between the trap and the underlying basement gneisses is clearly seen here and has an approximate altitude of 350 m. A total of 175 oriented block samples were collected from 29 sampling sites which were roughly equispaced over an altitude span of 583 m. The lowest sampling site was in a lava flow exposed in the bed of the river Godavari at an altitude of 344 m. The highest sampling sites (28 and 29) were located at an altitude of 927 m on the crest of the Mhaismal hill near Ellora (20°01'N 75°11'E). The altitudes were measured using an altimeter, with an accuracy of ± 2 m.

The terrain from Diglur up to Aurangabad is gently sloping with the average altitude increasing from 350 to 550 m. At Aurangabad one suddenly comes across hills ranging in altitudes up to 1000 m.

Sites 28 and 29 were located in a fresh rock exposure at the crest of the Mhaismal hill. At site 29, the compass needle behaved erratically when placed on the rock surface and finally swerved to a position which was about 90° from the local magnetic north. This suggested that the site was affected by a thunderbolt. Oriented samples were collected by first taking the compass bearing at a height of 1.5 m from the rock surface and then transferring the bearing on the sample. Site 28 is located in the same flow at a distance of about 10 m from site 29. Here the compass needle did not show any apparent local effect; this observation could be confirmed by comparing the bearing obtained here with the true

north direction engraved on a bench mark located in the compound of a nearby forest guest house. Oriented samples from sites 28 and 29 were collected mainly for studying the possible effect of lightning on their remanent magnetism.

LABORATORY MEASUREMENTS

Cores of 2.54 cm diameter were drilled out from each oriented sample block and sliced to obtain specimens having an approximate height of 2.5 cm. On an average, about four specimens were obtained from each block. The natural remanent magnetism (N.R.M.) of each specimen was measured on an astatic magnetometer having a sensitivity of $1 \cdot 10^{-6}$ Oe/mm scale deflection. The direction and intensity of remanent magnetism were calculated in each case.

In order to examine the primary character and stability of the R.M. of the specimens, a few pilot specimens from each site were subjected to alternating field (A.F.) demagnetisation. Each pilot specimen was demagnetised in a successively increasing alternating peak-field strength of 25, 50, 75, 100, 150, 200, 250 and 300 Oe. The direction and intensity of remanent magnetism, J, was measured after each stage of demagnetisation. The intensity of magnetism dropped to 15 % of the original value, J_0, on

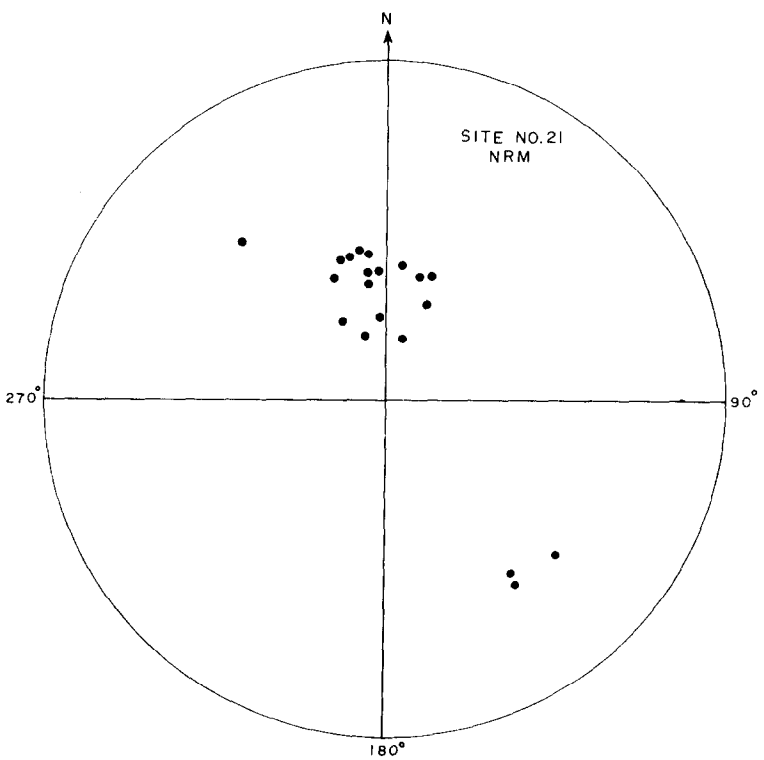


Fig.2. N.R.M.-directions for specimens from site 21 (equal-area projection).

demagnetisation in 300 Oe and the rate of decay of the remanent intensity flattened on demagnetisation at 50–100 Oe for specimens from most of the sites. The optimum alternating field value for batch-cleaning of all the specimens from each site was decided on studying the pattern of vector migration and intensity decay shown by the results obtained on the pilot specimens for the site. This procedure is illustrated in Fig.2–5 in the case of one site. Fig.2 is a plot on polar projection of the N.R.M. directions for specimens from site 21. Fig.3A illustrates the migration of the remanent-vector direction in the case of three pilot specimens from this site, on alternating field demagnetisation in 25, 50, 100, 150, 200, 250 and 300 Oe. It is clear from the figure that in the case of each specimen, the secondary components of magnetism have been removed on demagnetisation in 75 Oe, and that the remanent-magnetism vector has remained unchanged within statistical limits on cleaning at higher fields. Fig.3B shows the fractional J/J_0 loss in the intensity of magnetism of these specimens on demagnetisation in successively increasing values of alternating magnetic field. The slope of the curve is steep up to 75 Oe and flattens out at higher values of the peak-field. The peak-field of 75 Oe was therefore selected for batch-cleaning of the remaining specimens from site 21. Fig.4 is a plot on

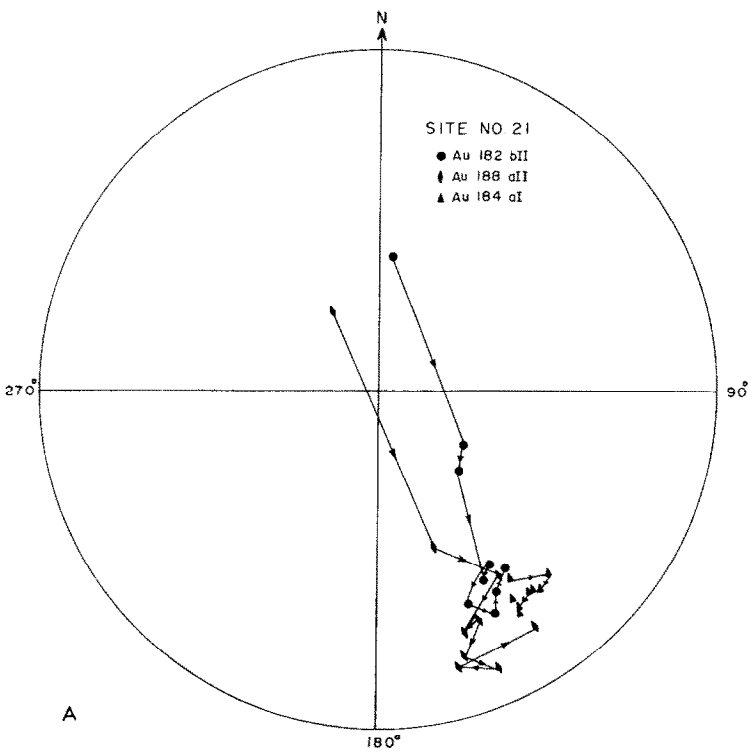


Fig.3A. The migration of remanent magnetism vectors for three pilot specimens from site 21, on successive demagnetisation in peak alternating fields of 25, 50, 75, 100, 150, 200, 250 and 300 Oe (equal-area projection).

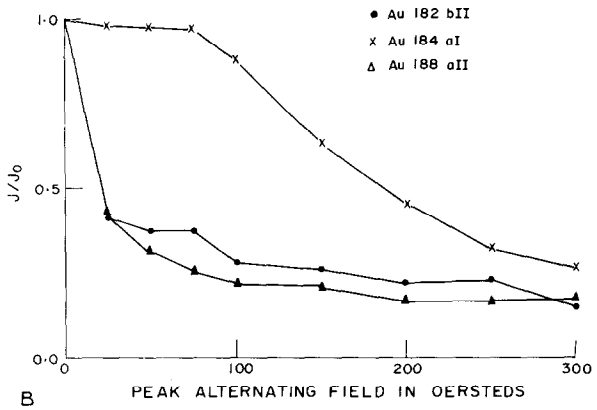


Fig.3B. Fractional loss (J/J_0) in the intensity of remanent magnetism of three pilot specimens from site 21 on demagnetisation in peak alternating fields as in Fig.3A.

polar projection of the remanent magnetic directions of all specimens from this site on demagnetisation in the peak-field up to 75 Oe. It is clear from Fig.2 and 4C that scatter in the directions has been considerably reduced and the effect of secondary components has been removed as a consequence of this treatment. Fig.5A–5G show the effect of A.F. cleaning on the remanent magnetic intensity of one pilot specimen from each site, excepting sites 5, 12, 28 and 29.

As explained earlier, site 29 was considered to possess anomalous magnetism attributable to lightning strike. The mean N.R.M. intensity value was found to be abnormally high ($73.0 \cdot 10^{-3}$ G/cm^3) as compared to the values for other sites (see Table I). Fig.6A shows the migration of the remanent-magnetism vector and Fig.6B shows the fractional loss in intensity of magnetism on A.F. cleaning in successively increasing peak-fields, in the case of three specimens from this site. It is clear from the figures that most of the magnetism is removed at a low value of the demagnetisation field (25 Oe) and although the N.R.M. direction is practically retained by each specimen, the directions do not show any tendency for grouping together. The specimens from site 28, which was located near site 29, behaved in a like manner, although no indication of their anomalous magnetisation was discernible at the time of sampling. The specimens from every sample from this site showed good grouping amongst themselves in N.R.M. and on A.F. cleaning, but the mean directions of the samples did not show any tendency of grouping together, although they all show reverse magnetisation.

In the case of sites 5 and 12, A.F. cleaning did not reduce the scatter in the N.R.M. directions, although the remanent direction of each specimen changed on demagnetisation. The results from these sites were not considered in further analysis. For these sites, the semi-angle of cone of confidence, α_{95} (Fisher, 1953), was greater than 15°, which was chosen as the upper limit for the acceptability of data from each site for further analysis.

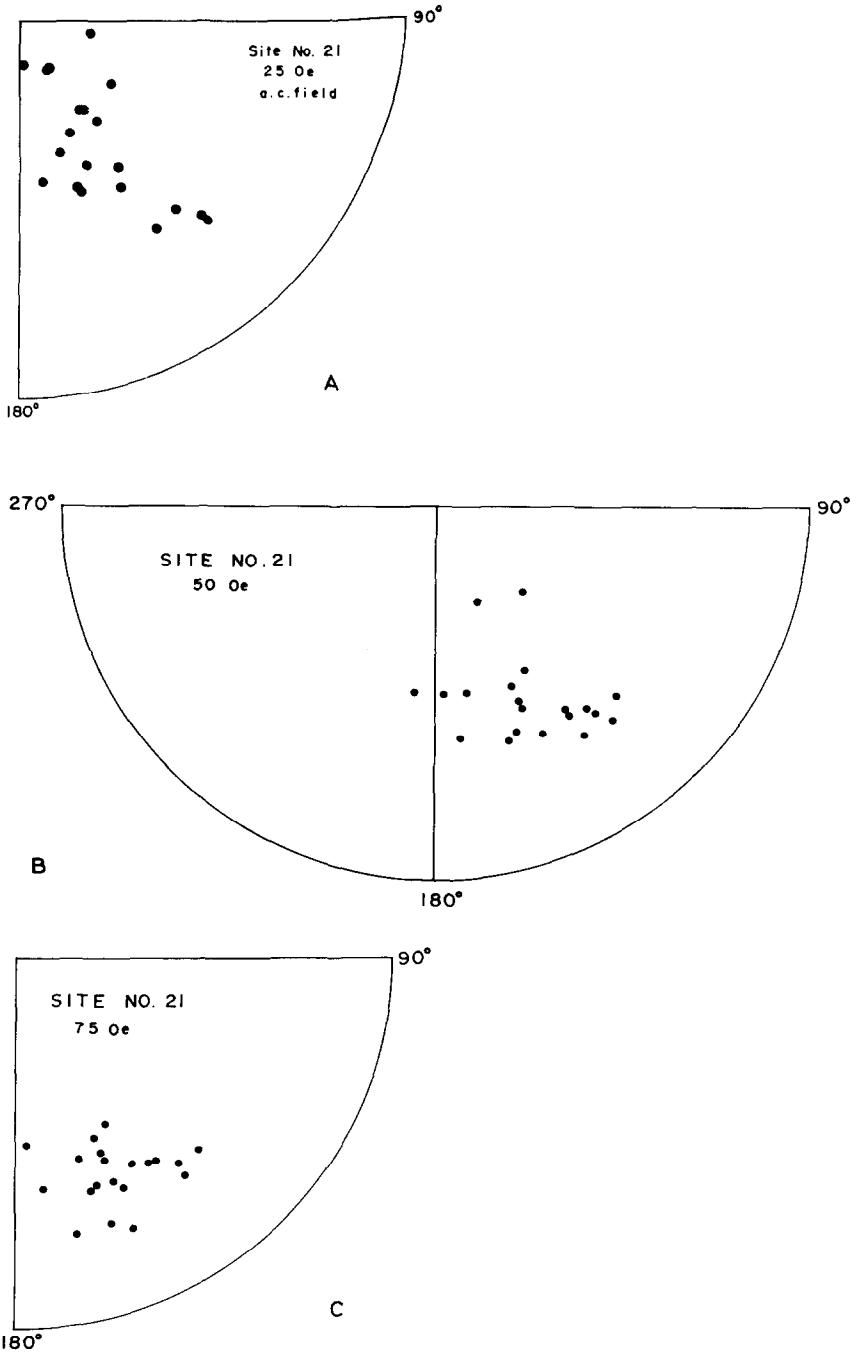
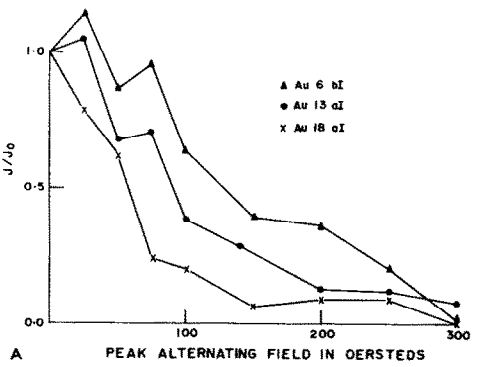
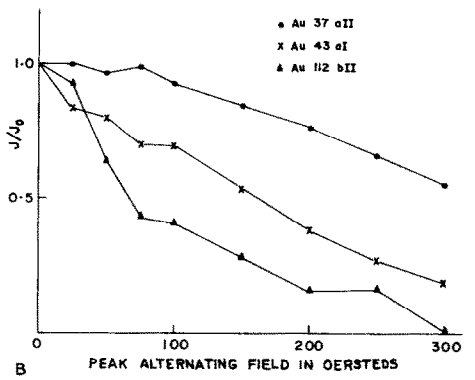
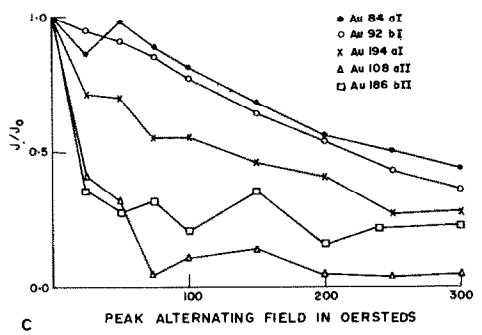
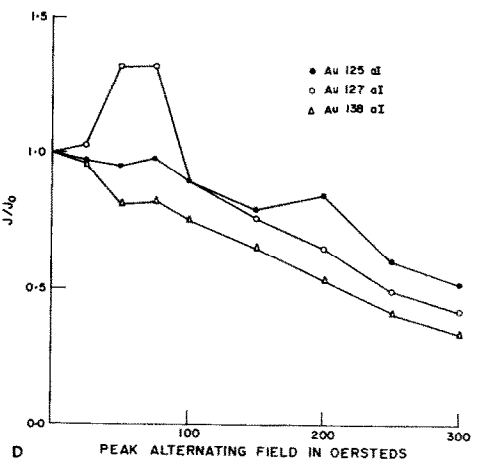


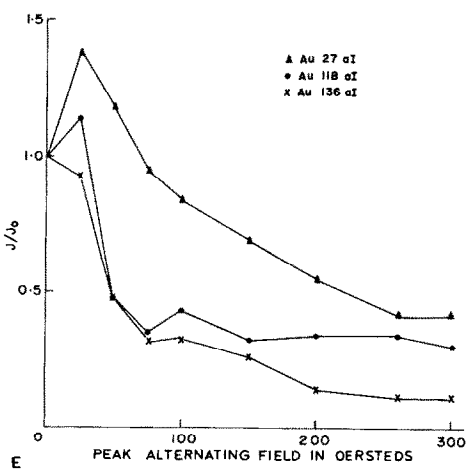
Fig.4. A plot on equal-area projection of the remanent magnetic directions of specimens from site 21 on demagnetisation in peak alternating fields of 25(4A), 50(4B) and 75(4C) Oe.











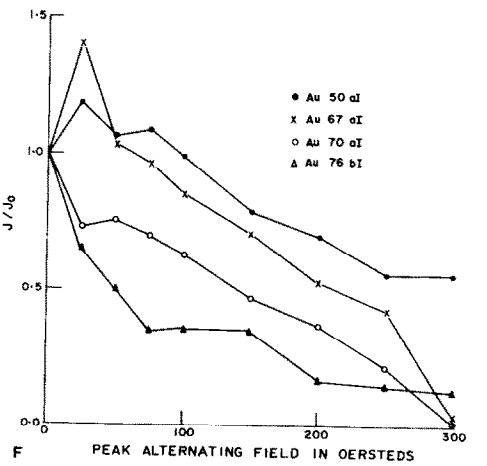


Fig.5A–F. For Fig.5G and legend see next page.

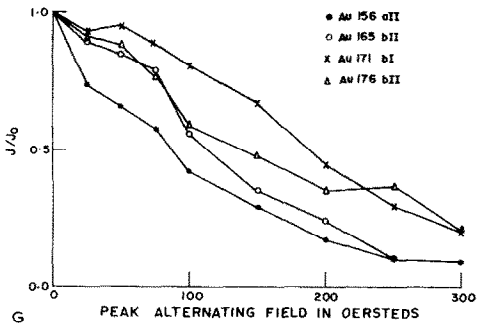


Fig.5. A–G. Fractional loss (J/J_0) in the intensity of remanent magnetism on successive demagnetisation in peak alternating fields of 25, 50, 75, 100, 150, 200, 250 and 300 Oe. (one specimen from each site).

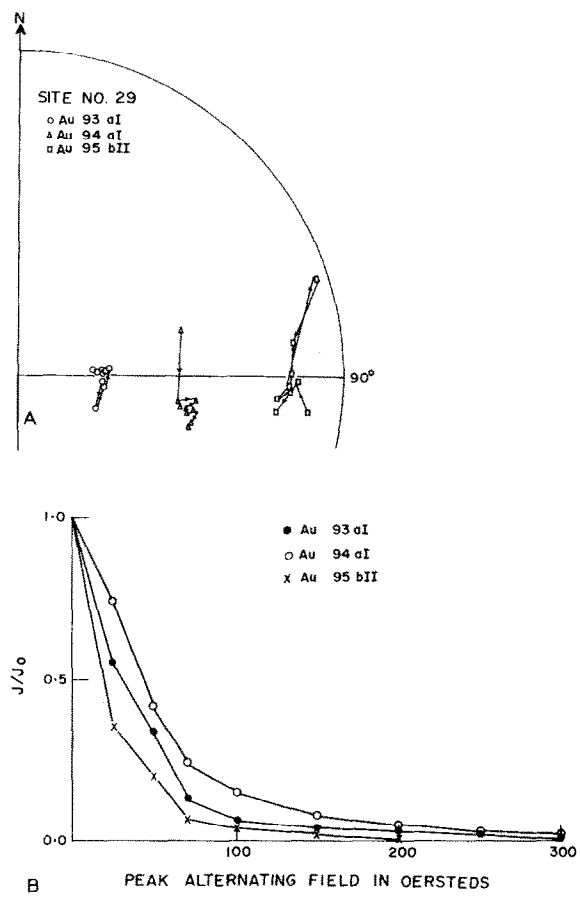


Fig.6. A. The migration of remanent directions of three specimens from site 29 on successive demagnetisation in peak alternating fields of 25, 50, 75, 100, 150, 200, 250 and 300 Oe. B. Fractional loss (J/J_0) in intensity of remanent magnetism of three specimens from site 29 on successive demagnetisation in peak alternating fields as in Fig.6A.

TABLE I

Palaeomagnetic results on Deccan trap from Aurangabad

Site	No. of samples	Elevation above M.S.L. (m)	Location of site		Mean N.R.M. intensity ($X10^{-3}$ G/cm^3)	Mean direction		R	K	α_{95} (°)	Pole position		Palaeolatitude of Nagpur (°S)
			north	east		D	I				°N	°W	
1	7	344	19°9'	77°21'	8.00	132	+61	6.91	67	7	14	72	52
2	4	381–384	19°43'	77°36'	25.72	150	+34	3.98	150	5	42	63	18
3	5	398	18°48'	77°33'	30.59	147	+53	4.99	400	3	29	71	33
4	6	450	19°36'	76°41'	5.66	112	+57	5.88	42	9	6	54	41
6	6	469	19°45'	76°6'	11.25	154	+46	5.96	125	5	37	75	27
7	3	503	19°48'	75°14'	11.96	155	+44	2.96	50	11	38	76	26
8	6	512	19°51'	75°54'	8.49	141	+42	5.88	42	9	32	62	25
9	5	518	19°50'	75°15'	5.74	137	+50	4.94	67	8	21	69	38
10	5	549	19°51'	75°16'	4.53	155	+27	4.92	50	9	48	67	14
11	6	555	19°51'	75°16'	6.39	149	+33	5.90	50	9	41	63	19
13	4	592	19°51'	75°16'	2.14	147	+48	3.90	30	13	32	71	30
14	6	591	20°1'	75°11'	8.09	151	+27	5.90	50	9	45	63	15
15	6	622	20°1'	75°11'	3.51	164	+27	5.93	71	7	52	79	14
16	19	631–646	20°	75°10'	7.51	174	+50	17.89	16	8	39	98	30
17	5	658	20°	75°10'	7.48	189	+69	4.97	133	5	18	111	50
18	5	683–686	20°	75°10'	7.92	181	+59	4.87	31	11	30	106	39
19	5	716	20°1'	75°11'	6.37	180	+60	4.80	20	14	29	105	40
20	5	744	20°	75°10'	8.32	133	+29	4.98	200	4	32	49	17
21	5	774	20°	75°10'	6.17	156	+35	4.97	133	5	44	72	23
22	3	793	20°2'	75°12'	9.74	149	+56	2.98	100	8	26	77	37
23	6	814	20°	75°10'	14.27	138	+56	5.79	24	11	21	80	43
24	4	826	20°2'	75°12'	12.24	148	+55	3.93	43	11	27	76	39
25	5	853	20°3'	75°11'	8.11	155	+57	4.92	50	9	6	59	40
26	5	884	20°5'	75°11'	9.46	135	+44	4.91	44	9	27	59	28
27	6	908	20°5'	75°11'	5.29	146	+54	5.90	50	9	27	74	36
Mean						150	+48			5.5	33	73	30

D = declination in degrees; I = inclination in degrees; R = length of resultant vector of n unit vectors; K = precision parameter; α_{95} = semi-angle of cone of confidence at 95 % confidence level.

ANALYSIS OF THE DATA

Samples, rather than specimens, were regarded as the primary units of data for statistical analysis. The directions of specimens from each block were always close to one another and may be subject to a common orientation error. The reasonable agreement in the directions of specimens from a block-sample suggested that a simple arithmatic mean was justifiable, since the main variation was found to be between the samples rather than amongst the specimens from a single sample. The mean directions obtained for each site according to Fisher's (1953) procedure were then used to obtain the mean stable direction of remanent magnetism for the entire lava sequence.

Assuming that the landmass was drifting, the mean direction of each flow was used in calculating the palaeo-latitude of a centrally located town (Nagpur: 21°N 79°E), at the time of its magnetisation. The same mean directions were used in calculating the position (Hospers, 1955) of the geomagnetic pole at the time of magnetisation of each flow.

Table I gives the mean stable remanent direction and N.R.M. intensity value for each site. Also given are the corresponding palaeo-latitude, pole position, the statistical parameter of precision K, semi-angle of cone of confidence at 95 % level (Fisher, 1953) and other details such as location, altitude and number of samples. Fig.7 shows the variation with altitude in declination and inclination of the mean stable remanent direction obtained for each site. In Fig.8 are plotted virtual geomagnetic pole positions for all sites along with the mean pole position and its oval of confidence.

Fig.9 is a plot showing the mean pole position for the cumulative data for upper (normally magnetised) and lower (reversely magnetised) Deccan trap. These mean pole-

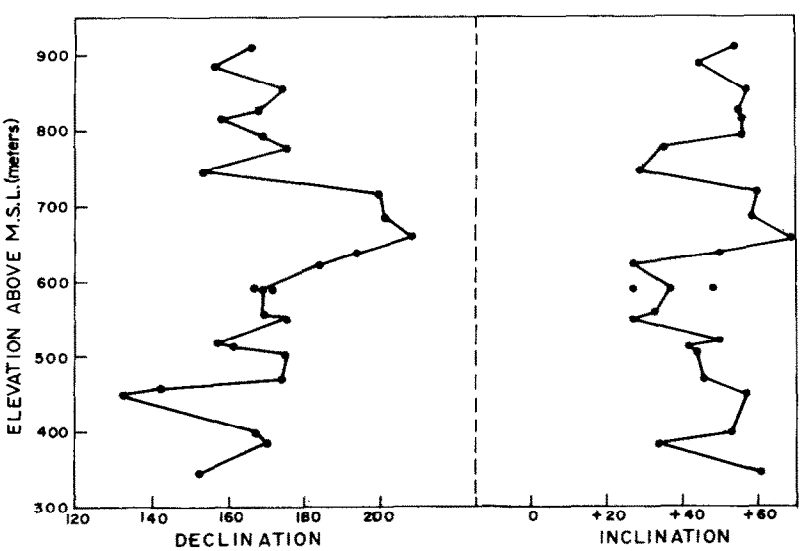


Fig.7. Variation in declination and inclination with altitude of sampling sites.

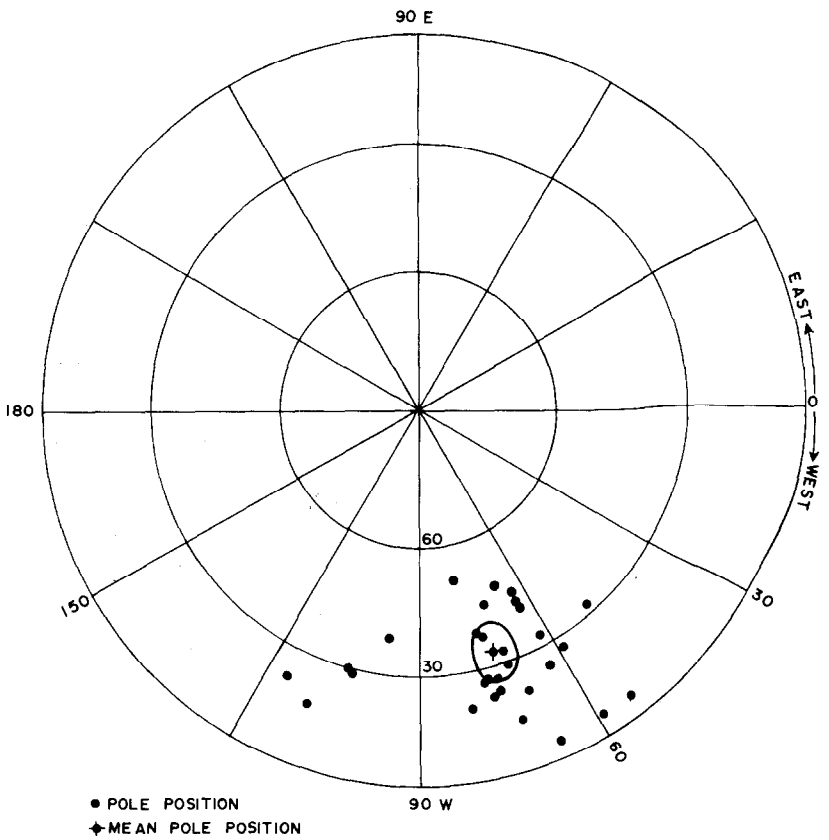


Fig.8. Virtual geomagnetic pole positions for Deccan trap lava flows from the Aurangabad region (equal-area projection).

position values have been reproduced from Athavale et al. (1970). Also shown in this figure is the mean pole-position value obtained for the trap sequence in the Aurangabad region.

DISCUSSION

The results presented in Table I show that all the lava flows in the Aurangabad region between altitude range 344–927 m are reversely magnetised (azimuth nearer to 180° and inclination downwards). In our study of the Ellora section, covering the altitude span of 591–927 m, we could not locate any polarity transition, indication for which was given in the earlier preliminary studies reported by Bhimasankaram and Pal (1968).

The Deccan trap has been palaeomagnetically classified into two groups: (1) lower (reversed) and (2) upper (normal) with the demarcation line running along an altitude of 600 ± 60 m. Two cases of apparent departure from this classification have been reported

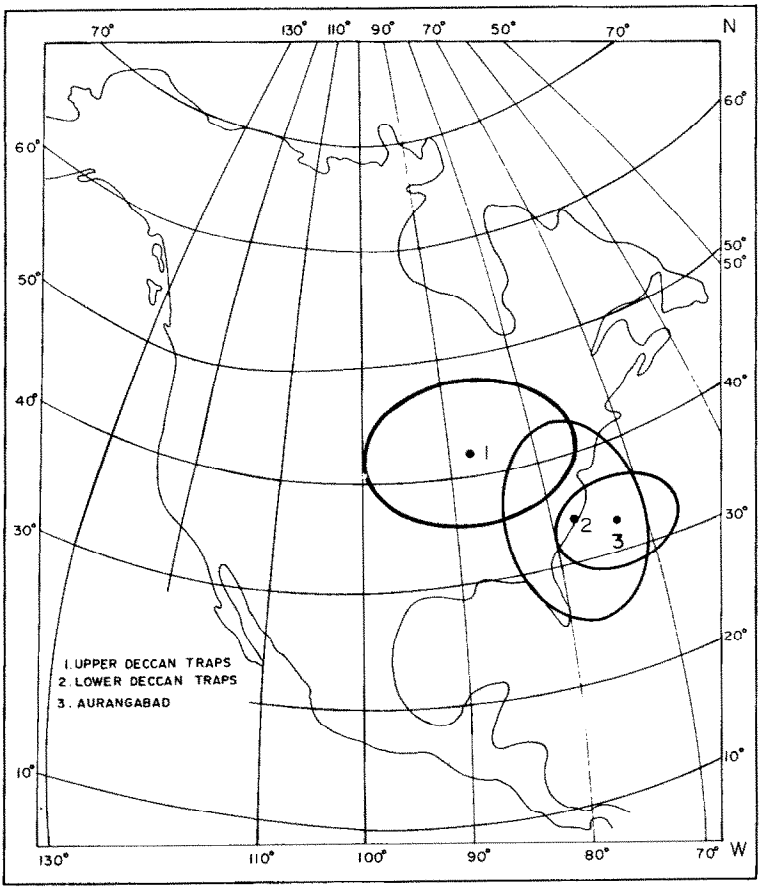


Fig.9. Mean palaeomagnetic pole positions for the upper and lower Deccan trap along with the mean pole position for the Deccan trap of the Aurangabad region.

so far. Sahasrabudhe (1963) studied the remanent magnetism of Deccan trap rocks exposed at Bombay at sea-level and found them to be normally magnetised. On the basis of field evidence, earlier geologists had suggested (Pascoe, 1964) that the great scarp running parallel to the west coast in the Deccan trap country is due to a boundary fault, with the downthrow on the western side. According to Sahasrabudhe, this appears to be the most plausible explanation for occurrence of normally polarised rocks at Bombay near sea-level while they are present at an altitude of more than 600 m in the scarp sequences.

The other case is reported by Athavale (1969, 1970), who studied a 300 m sequence of Deccan trap flows exposed in the northeast corner, between Amarkantak and Dindori (Fig.1). He found that the polarity transition from reverse to normal appears at an altitude of about 975 m. On the basis of geological, geomorphological and geophysical evidence, he was able to establish that the polarity transition located at Amarkantak at an altitude

of 975 m was the same as the one recorded near Jabalpur at an altitude of about 625 m (Verma and Pullaiah, 1968), and that the difference in the altitudes is due to post-trap uplift of the Amarkantak block by about 350 m. These two examples suggest that the Deccan trap has been subjected to tectonic disturbances causing block-rejuvenation and subsidence. The present topography of the Deccan trap therefore does not appear to be the consequence of erosion processes alone.

In the light of the discussion above and the background given in the introduction to this paper, the absence of the reverse to normal polarity transition in the palaeomagnetic results on the Deccan trap from the Aurangabad region could be explained through one of the following three possible interpretations:

(1) The reversely magnetised lava flows in the Diglur–Aurangabad sector are all older than those on the western coast where a transition from reversely magnetised to normally magnetised flows occurs at an altitude of 600 ± 60 m.

(2) The lava flows of the two regions may be of the same age but may have different eruption history. In other words, in the same span of time, much more lava poured out in the Diglur–Aurangabad region to form a pile having a thickness greater than that of the reversed rocks on the western coast.

(3) The Aurangabad region has been uplifted at least by about 325 m as a result of post-trap movements.

These three possibilities arise from a common basic assumption that the reverse to normal polarity transition found at an altitude of 600 m at several locations in the Deccan trap, can be used as a stratigraphic marker. The validity of this assumption needs to be examined first.

Wellman and McElhinny (1970) have reported whole-rock K/Ar ages on four samples of the Deccan trap. On the basis of this data, they estimate the average age of the Deccan trap at around 60 m.y. and the span of the eruption activity as 5 m.y. Heirtzler et al. (1968) have compared the magnetic anomaly data obtained in marine surveys with the palaeomagnetic record of reversals of the earth's magnetic field and they have prepared a time-scale going back 80 m.y. and recording the various polarity epochs which have taken place over this period. They make a frequency analysis for the interval between two polarity transitions and show that the frequency of reversals reduces with increasing age. Near about 60 m.y., they estimate the frequency of reversals as 1.5 events for every 5 m.y. and 1 event for every 2 m.y. Accepting the estimate of Wellman and McElhinny (1970) for the span of Deccan trap activity as 5 m.y. and assuming that the time-scale given by Heirtzler et al. (1968) is universally applicable, it would appear that at the most two reversals of the earth's magnetic field could have been recorded in the Deccan trap sequence whose average age is 60 m.y. Recently, Pal et al. (1971) have reported the occurrence of normally magnetised flows below reversely magnetised flows at an altitude of 360 m, in a sequence from the Malawa region constituting the northern sector of the Deccan trap country. They state that this result has been obtained after intensive alternating field demagnetisation tests. This then appears to be the second (earlier) reversal recorded in the Deccan trap and expected on the basis of the frequency analysis

by Heirtzler et al. (1968). This reversal from normal to reverse is most probably the same as the one recorded in the Tirupati Sandstones (Verma and Pullaiah, 1967) which are considered as coeval with the lower Deccan trap.

Three possibilities

(1) Although Wellman and McElhinny (1970) estimate the span of the Deccan trap activity as 5 m.y., the actual period could be much less. One of the dates reported by them is on a sample from Zahirabad (17°40'N 77°36'E), which is located near the eastern margin and can be considered as representative of the flows studied in this paper. The age of this basalt was found to be 59.0 ± 1.1 m.y. The other date of our interest is that for the basalt sample coming from the base of Mt. Pavagarh (22°28'N 73°31'E), which is capped by rhyolites and is considered as representing the late phase in Deccan trap igneous activity (Wadia, 1961). The mean of two values reported for this Pavagarh sample comes to 61.7 ± 1.8 m.y. The date for the Pavagarh Basalts, which are probably the youngest basic rocks in the Deccan trap, thus overlaps the date for the trap located hundreds of kilometers away at Zahirabad on the eastern margin. This radiometric data, although quite meagre, tends to suggest that the eruption activity was very fast and the trap in different regions could be considered as contemporaneous. The time span involved appears to be so small that the resolution of age difference in the lava flows of various regions in the Deccan trap does not seem to be feasible with the present limitations of radio-chronometry.

(2) Sahasrabudhe (1963) has reported the occurrence of reversely magnetised rocks at sea-level near Kalyan (19°30'N 73°7'E) on the west coast. From this one can estimate the thickness of the reversely magnetised trap on the western coast up to the polarity transition level as 600 m. This thickness compares very well with the thickness of 583 m studied by us in the Aurangabad region, although the possibility of the existence in the geological past of some more flows (presently eroded) above the 927 m level, up to the reverse to normal polarity transition level in this region, cannot be ruled out.

Another parameter which can give some idea of the rate of eruption of lava flows, is the frequency of intertrappean beds. These beds in the Deccan trap are usually very thin (1–3 m) and they have been formed through processes operating during the quiescent phases of the volcanic activity. According to Wadia (1961), these intertrappean beds are fairly regularly distributed throughout the lower and upper trap. This observation suggests that the frequency of eruption of lava flows was roughly the same on both the eastern and western sides throughout the period covered by the Deccan trap activity. It would thus appear that although the second possibility cannot be entirely ruled out, its probability does not seem to be so high as to preclude the examination of other possible interpretations.

(3) The third possibility, which suggests an uplift of the Aurangabad region, seems more likely in view of the known examples of similar movements causing block uplift and subsidence in the case of Amarkantak (Athavale, 1970) and the Bombay trap (Sahasrabudhe, 1963). Several cases of block-faulting have been recognised by geologists

in the case of the Deccan trap flows which constitute the E–W trending Satpura Mountain system which is located on the north side of the Aurangabad–Ellora hills (Fermor, 1916, Pascoe, 1964).

Examination of the Bouguer anomaly map of India prepared by Qureshy (1964) shows that a major portion of the Deccan trap is associated with a gravity low. The Bouguer anomaly pattern near Aurangabad shows a low of –70 mgal flanked by relatively higher enclosures of –60 mgal on the western and southwestern sides and a markedly positive gradient on the northern and northeastern sides. The Aurangabad region falls south of the E–W trending Satpura chain which shows a broad high, having an average Airy-Heiskanen anomaly of +6 mgal. This isostatic anomaly has been interpreted by Qureshy (1964) in terms of a horst-like nature of the Satpura mountain chains and an upwarping of the subcrust by approximately 240 m.

Qureshy (1971) has recently studied the relation of gravity to elevation, in India. According to him, a regression study of the Bouguer anomaly versus elevation, made by grouping the gravity data according to tectonic and geologic units, shows that the Bouguer anomaly decreases rather slowly with the increase in elevation, which suggests that most of the topographic "highs" could be interpreted in terms of the rejuvenation of blocks of the Archean shield. Since the Deccan traps rest unconformably on the basement of Archean rocks, the occurrence of similar rejuvenation of discrete shield blocks underneath them can be considered as very likely.

CONCLUSIONS

(1) The Deccan trap flows exposed on the eastern side of the trap country and in the hills around Aurangabad between altitudes 344–927 m are all reversely magnetised.

(2) Examination of the available palaeomagnetic data on the Deccan trap suggests that only two polarity transitions are likely to have occurred during the Deccan trap volcanic activity whose span has been estimated up to 5 m.y. One of these polarity transitions is recorded near the base of the trap and is a normal to reverse transition. The other, a reverse to normal transition, appears at an intermediate level at many localities in the Deccan trap and can be considered as a useful stratigraphic marker for correlation of flows from different regions of the Deccan trap and for understanding post-trap tectonic movements.

(3) Palaeomagnetic results for the lava flows in the Aurangabad region suggest that this area might have been uplifted by at least about 325 m as a consequence of block-faulting.

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